

Lean port sheets

Rev F0 | 2026-10-04 | Schematics authored by Chinmoy Bhuyan | D-044 lean ports A (PV) and B (battery) + insulation monitor, BUS- referenced

01 Supplies and commands

Live 24 V / 5 V / 3.3 V, BUS- tie, commands

File: sheets/01_lean_io.kicad_sch

02 Port A power path (pv)

Varistor Y + monitor

File: sheets/02_leanA_power.kicad_sch

03 Port A sensing and interlocks

References, BUS- shunt amplifier + hold-off / OC windows, dividers, dV amplifier, polarity and dV interlocks

File: sheets/03_leanA_ctrl.kicad_sch

04 Port A coil drives

Contactor: interlock logic, fail-safe high side, economiser; passive bank bleeder

File: sheets/04_leanA_drive.kicad_sch

05 Port B power path (battery)

Varistor Y + monitor, aR fuses

File: sheets/05_leanB_power.kicad_sch

06 Port B sensing and interlocks

References, BUS- shunt amplifier + hold-off / OC windows, dividers, dV amplifier, polarity and dV interlocks

File: sheets/06_leanB_ctrl.kicad_sch

07 Port B coil drives

Contactor + precharge relay: interlock logic, fail-safe high side, economiser; passive bank bleeder

File: sheets/07_leanB_drive.kicad_sch

08 Insulation monitor

PE divider, two switched 992 k strings through CA-IS3417WT

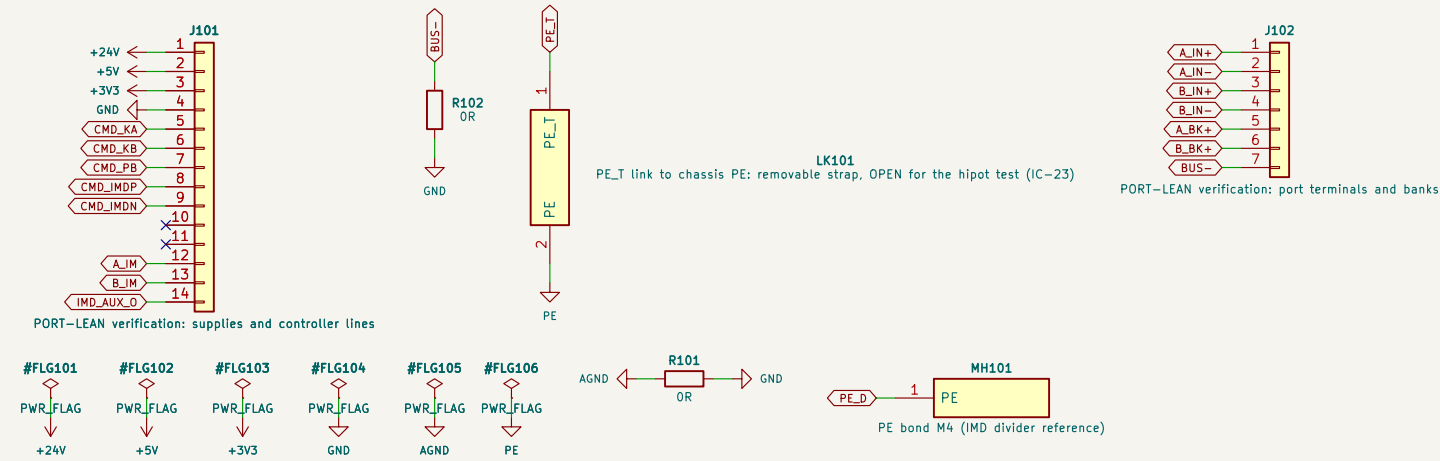
File: sheets/08_lean_imd.kicad_sch

Verification build of the reusable lean-port functions of gen/port.py (lean_port, varistor_y, lean_contactor, lean_precharge, lean_bleeder, lean_dividers, lean_shunt, lean_interlock, lean_imd). The product sheets live on the power board.

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'
Sheet 1
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: / File: PORT-LEAN-HOLD.kicad_sch		
Title: Lean port sheets – Cover / Hierarchy		
Size: A3	Date: 2026-10-04	Rev: F0
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SUPPLIES AND COMMANDS



On the power board these come from the AUX-T1 live winding and the controller.

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'
Sheet 2
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /01 Supplies and commands/
File: 01_lean_io.kicad_sch

Title: Lean port sheets – Supplies and commands

Size: A3	Date: 2026-10-04	Rev: F0
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PORT A (PV SOURCE), D-044 LEAN PORT

Terminal side A_{LN}+ / A_{LN}-, bank A_{BK}+. No fuse, no precharge; array lsc <= 1.25 x rating.
 Contactor opens <= 0.97 kA, held above [port_spec lean]. Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side A_{IN+} / A_{IN-}, bank A_{BK+}. No fuse, no precharge: array lsc <= 1.25 x rating.
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

ON-BOARD SURGE PROTECTION: VARISTOR Y (IC-01, IC-12)

TV125751 x 3 (615 VDC, thermal link + monitor lead); MOV1 +/M, MOV2 -/M, MOV3 M/PE-1, Up.eff 3793 V pole-PE and pole-pole of in 3 kA (calc.; <= 4.0 VV -> reinforced 6 kV / basic 4 kV, Dr=032); I_max 25 kA, Branch fuses ETI CH14x51 gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line); RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

TVT25751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE-T. Up,eff 3793 V pole-PE and pole-pole at in 3 kA (calc., ≤ 4.0 kV $\rightarrow$ reinforced 6 kV / basic 4 kV, D-032); I_{max} 25 kA. Branch fuses ETI CH14x51 gPV 36 A 30 k (L/R 2 ms). Monitor loop through both fuses and all three links $\rightarrow$ CNY65B $\rightarrow$ RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open $\rightarrow$ CTRL inhibits/stops.

Sheet: /02 Port A power path (pv)/
File: 02_leanA_power.kicad_sch

Title: Lean port sheets – Port A power path (pv)		
Size: A2	Date: 2026-10-04	Rev: F0
KiCad E.D.A. 10.0.6		Id: 3/9

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'

Sheet 4

Schematics authored by Chinmoy Bhuyan

Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /03 Port A sensing and interlocks/

File: 03_leanA_ctrl.kicad_sch

Title: Lean port sheets – Port A sensing and interlocks

Size: A3

Date: 2026-10-04

Rev: F0

KiCad E.D.A. 10.0.6

Id: 4/9

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'		
Sheet 5		
Schematics authored by Chinmoy Bhuyan		
Live side referenced to BUS-; basic HV-PE; functional inside the live side		
Sheet: /04 Port A coil drives/ File: 04_leanA_drive.kicad_sch		
Title: Lean port sheets – Port A coil drives		
Size: A3	Date: 2026-10-04	Rev: F0
KiCad E.D.A. 10.0.6	Id: 5/9	

PORT B (BATTERY SOURCE), D-044 LEAN PORT

Terminal slide B₁/N₁ / B₂/N₂ - bank B₁/N₁, HP1501 250 A air in both poles (breaks 1.25-50 AA only).
 Contactor opens <= 0.97 A, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side B_{JN+} / B_{JN-}, bank B_{BK+}, HPIE501 250 A aR in both poles (breaks 1.25–50 kA only).
Circuit breaker opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

ON-BOARD SURGE PROTECTION: VARISTOR Y (IC-01, IC-12)

The diagram illustrates the on-board surge protection circuit for Varistor Y, showing two main sections: IC-01 and IC-12.

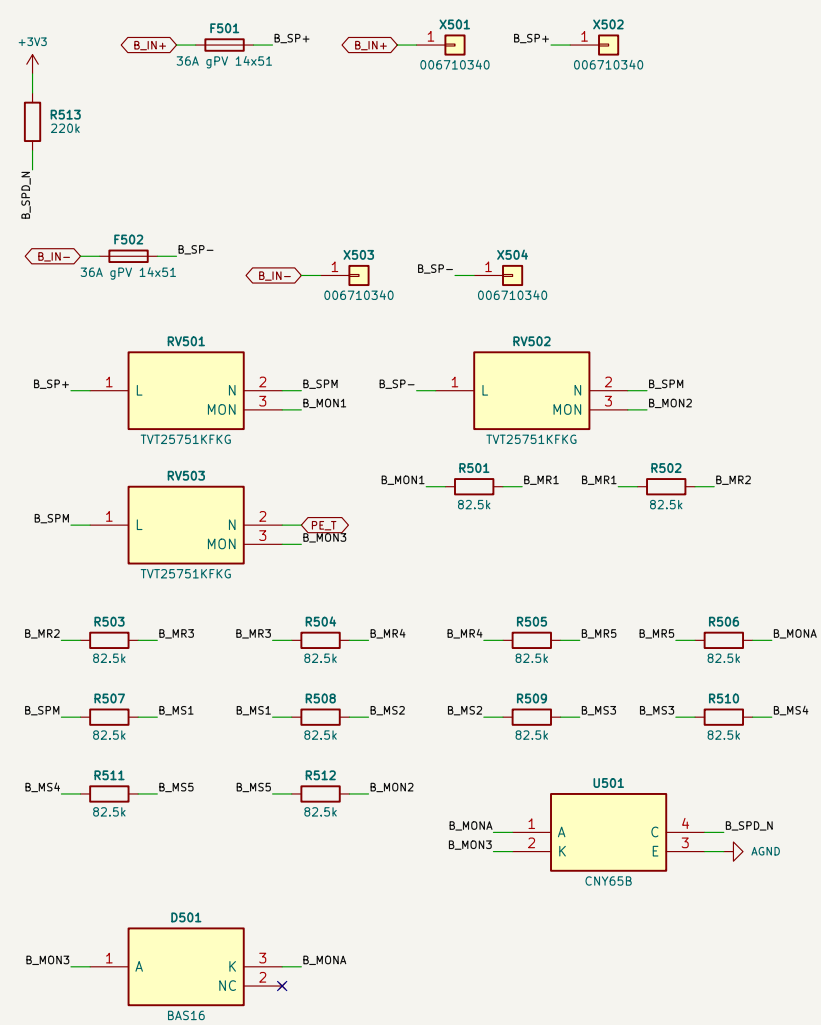
IC-01 Section:

- Top Circuit:** A varistor (RV01) is connected in series with a 36A gPV 1x51 component. The circuit is labeled with +VS, 00B71C24D, and 00B71C24D.
- Bottom Circuit:** A varistor (RV02) is connected in series with a 36A gPV 1x51 component. The circuit is labeled with 00B71C24D, 00B71C24D, and 00B71C24D.

IC-12 Section:

- Top Circuit:** A varistor (RV01) is connected in series with a 36A gPV 1x51 component. The circuit is labeled with 00B71C24D, 00B71C24D, and 00B71C24D.
- Bottom Circuit:** A varistor (RV02) is connected in series with a 36A gPV 1x51 component. The circuit is labeled with 00B71C24D, 00B71C24D, and 00B71C24D.

The diagram also includes a detailed view of the varistor (RV01) and its connection to the circuit, showing the varistor's internal structure and the connection points for the surge protection circuit.



TV125751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE_T. Up,eff 3793 V pole-PE and pole-pole at 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D-032); Imax 25 kA, Branch fuses ETI CH1x51 gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNY65B

FUSES (BATTERY-TYPE PORT)

HPE501/000B100-250 in both poles: breaks 1.25-50 kA only (partial range)
the part OC trip and the contactor keep it out of 250 A-1.25 kA.

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'
Sheet 7
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /06 Port B sensing and interlocks/
File: 06_leanB_ctrl.kicad_sch

Title: Lean port sheets – Port B sensing and interlocks

Size: A3

Date: 2026-10-04

Rev: F0

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Id: 7/9

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'

Sheet 8

Schematics authored by Chinmoy Bhuyan

Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /07 Port B coil drives/

File: 07_leanB_drive.kicad_sch

Title: Lean port sheets – Port B coil drives

Size: A3

Date: 2026-10-04

Rev: F0

KiCad E.D.A. 10.0.6

Id: 8/9

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'

Sheet 9

Schematics authored by Chinmoy Bhuyan

Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /08 Insulation monitor/

File: 08_lean_imd.kicad_sch

Title: Lean port sheets – Insulation monitor

Size: A3

Date: 2026-10-04

Rev: F0

KiCad E.D.A. 10.0.6

Id: 9/9